Even sign changes and the rank of a string

Human Compression project: F006 recovery

Author working proof; distinct review pending

Put $E_n = \{v \in \mathbb{F}_2^n : \sum_i v_i = 0\}$ and $W_n = E_n \rtimes S_n$, with the natural permutation action on coordinates. We retain the classical identification $W(D_n) = W_n$. In the signed action on $\{\pm 1, \ldots, \pm n\}$, $v_i = 1$ exchanges i and $-i$. The two retained symmetric-group inputs are stated precisely below. An independent family means that no member belongs to the subgroup generated by the others; it need not have the intersection property.

Lemma 1 (The module and relative generation)**.** *For $n \geq 3$, every S_n-invariant subspace of E_n is either zero, the constant line (when n is even), or E_n. If an independent family generates W_n, and L is any inclusion-minimal subfamily whose images generate S_n, at most one member lies outside L.*

Proof. If an invariant subspace contains a nonconstant vector v, choose i, j with $v_i \neq v_j$. Then $v + (i\ j)v = e_i + e_j$ belongs to it. Conjugating gives every vector of weight two, which spans E_n. The only nonzero constant vector is $\lambda = (1, \ldots, 1)$, and it belongs to E_n exactly when n is even.

For each generator g outside L, choose a word w_g in L with the same image in S_n and set $v_g = g w_g^{-1} \in E_n$. The new list $L, (v_g)$ generates the same group. It remains independent with respect to the entries v_g: if one belonged to the subgroup generated by all other entries, then so would the corresponding g. Let K_g be generated by L and all v_h with $h \neq g$. Its image in S_n is full. Conjugation of E_n by K_g is consequently the full S_n action, since E_n is abelian. If v_g belonged to the S_n-span of the other v_h, it would belong to K_g, a contradiction. Thus the v_g are independent under generation of S_n-submodules. There can be at most one: a nonconstant one spans all of E_n, and two nonzero constant ones would be equal. Zero entries are impossible. □

This is the normal-module normalization already used in the project's common theory *Forest coordinates and relative generation* and its transitive-rank supplement. The proof is repeated here to make the stronger consequence for this particular module explicit.

The retained Whiston bound says that an independent generating family of S_n has at most $n-1$ members. The acquired extremal string result says that an independent generating family of $n-1$ involutions in S_n which satisfies the string commutations is a Coxeter path, up to an automorphism of S_n, for $n \geq 5$. For $n \geq 7$ it follows from the common Cameron–Cara classification: the sign modification of a tree has a generator of order three, and the line graph of an ordinary transposition tree satisfying the string condition must be a path. The acquired complete S_5, S_6 certificate gives the two remaining degrees, including the outer image of the path in S_6. Neither input is inferred from a projection having the intersection property.

Theorem 2. *Every independent generating family of W_n, $n \geq 5$, has at most n members. If n is even, every such family of involutions satisfying the string commutations has at most $n-1$ members. In particular these bounds apply to every string C-marking, connected or disconnected.*

Proof. Choose L as in Lemma 1. Its images are an independent generating family of S_n, so $|L| \leq n-1$. The lemma gives total length at most $|L| + 1 \leq n$.

Suppose n is even and a string family has length n. Equality forces $|L| = n-1$. Its images are nonidentity involutions and retain the string commutations after the other position is deleted.

By the acquired extremal result they form a Coxeter path. Every neighboring pair in this path fails to commute. Thus their original positions were consecutive as well; the one omitted position is an endpoint of the original string. Reverse that string if needed and call its first, omitted generator x.

The element x commutes with the last $n-2$ members of L. Their images generate an endpoint-omission subgroup $Q \cong S_{n-1}$ of the Coxeter path. Its centralizer in S_n is trivial, so x has trivial image and is a nonzero vector in E_n. Commutation says that this vector is constant on every Q-orbit. For the natural path these orbits have sizes $1, n-1$. Since $n-1$ is odd, even weight makes the two constants equal. For the outer S_6 path the endpoint subgroup is transitive on six letters, so its invariant vectors are also constant. These assertions for the two degree-six markings follow directly from the explicit acquired permutation representatives. Hence $x = \lambda$ is central in W_n.

Put $H = \langle L \rangle$. Since H maps onto S_n, $H \cap E_n$ is an S_n-submodule. Independence excludes $\lambda \in H$, while Lemma 1 shows that every nonzero submodule contains λ when n is even. Therefore $H \cap E_n = 0$. But $W_n = \langle H, \lambda \rangle = H\langle\lambda\rangle$ would then have kernel $\langle\lambda\rangle$ over S_n, whereas its kernel E_n has dimension $n-1 > 1$. This contradiction proves the second assertion. □

Theorem 2 replaces the upper-rank argument in Chen–Rowley, *Abstract Regular Polytopes of Finite Irreducible Coxeter Groups*, EJC 32(4) (2025), P4.18, Theorem 12 and the upper bounds in Theorems 16 and 21. It uses no transitive half-degree theorem, and eliminates the source's separate case in which a minimal quotient family has length $n-2$: the module normalization rules out that case at once. Its retained Whiston and extremal inputs, including the complete small certificate, remain mathematical dependencies.

Scope. This working note proves the upper bounds. Construction of every rank, the remaining exceptional census rows, and distinct review remain separate obligations. No whole-original, open-question, formalization or numerical compression credit is claimed here.